

Doom PWAD QA Report

Run b9b52f83-73ba-4feb-9b8d-ded96e138dc6 | Map MAP01 | Outcome stuck

Executive Summary

This is an autonomous QA test of MAP01 using a lockstep LLM/MCP test harness. The test was designed to evaluate navigation, combat engagement, resource balance, and secret discovery within the map. The run concluded with a 'stuck' outcome after 331 seconds, failing to complete the map due to persistent navigation and resource-related defects.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL

overall_verdict: FAIL

- The run ended with outcome 'stuck' after visiting 21 coarse position cluster(s).
- 0 kill(s) recorded against 12 enemy/enemies that spawn at difficulty 3.
- Selected-difficulty health_ratio=0.0000, ammo_ratio=0.0833.
- 0 secret(s) found.

No technical problems encountered during this test run.

Test Item And Environment

The test evaluated MAP01 using the Freedoom2 IWAD at skill level 3. The map contains 12 enemies (Imps and Demons), 12 items, 8 weapons (including Chainsaw, Shotgun, and Plasma Rifle), 4 secret sectors, and measures 3840x3840 units. The map demonstrates high static complexity with several inaccessible weapon placements.

The test environment utilized ViZDoom, controlled by the gemini-3.1-flash-lite LLM model via an MCP tool interface. The run was capped at 600 ticks, with a focus on autonomous navigation and combat performance.

Metric	Value	Metric	Value
Status	completed	Duration	331
Actions	42	LLM calls	42
Final HP	94	Kills	0
Position samples	341	Movement units	7937.7
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/b9b52f83-73ba-4feb-9b8d-ded96e138dc6.mp4		

Recording And Agent Quality

Recording	Value	Agent progress	Value
Status	ok	Progress score	18
Frames	253 total / 246 unique	Meaningful progress	6
Video shape	640 x 480 @ 15.0 FPS	Completed objects	6
Gameplay time	16.829s, 589 tic(s)	Quality flags	warning

Quality warnings: Object 34 was already reached or collected in this run, so I will not target it again.

The requested action repeats a recent no-progress signature (explore), so I am switching tactics.

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Combat against target 7 stopped with out_of_ammo.

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Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	12	12	0 hidden by skill/multiplayer flags
Items	12	12	0 hidden by skill/multiplayer flags
Secrets	4	4	Sector special 9
Geometry	208 linedefs	30 sectors	3840 x 3840 units

Selected-difficulty estimate: hard. Health ratio: 0.0. Ammo ratio: 0.0833.

Coverage Evaluation

Coverage was limited to a small, initial region of the map. The automated playthrough explored fewer than 30% of the available sectors and engaged 0 of the 12 spawned enemies due to severe resource starvation. The coverage was insufficient to evaluate the map's full gameplay loop.

Objectives Covered

- **Map launch and initial state capture:** Run status: completed.
- **Director decision logging:** 42 decision event(s) recorded.
- **Movement telemetry:** 341 position sample(s) across 21 cluster(s).
- **Gameplay recording:** /media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/b9b52f83-73ba-4feb-9b8d-ded96e138dc6.mp4

Objectives Omitted

- **Navigate the entire map:** Navigation was stalled at tick 126 and repeatedly failed thereafter.
- **Engage all enemy types:** Total kills remained at 0; automated playthrough did not acquire ammo.
- **Find secrets:** 0 secrets found; automated playthrough did not access relevant areas.
- **Stress-test geometry:** Automated playthrough confirmed unreachable weapon placements due to collision barriers.
- **Document resource balance:** Resource balance documented as a critical failure (ammo starvation).

Uncovered attributes: The playthrough was unable to test map-specific scripting, teleporters, or any progression-gating switches. The automated playthrough observed unexpected collision barriers that blocked access to all provided weaponry.

Defects And Risks

Five defects were detected during this run, including one critical navigation softlock, three major issues (resource starvation, progression, and geometry), and one minor secret discovery issue. The most impactful defect was the navigation softlock which completely halted map traversal.

Patterns: Defects were concentrated in the automated playthrough's inability to navigate around collision barriers, leading to a recurring pattern of ammo starvation and eventual total progress stasis.

Severity	Priority	Type	Tick	Description
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1	1	softlock_navigation	126	Run stalled after repeated stuck decisions The run produced repeated stuck events with no meaningful progress. This usually indicates blocked navigation, unreachable objectives, or an automated control loop that should be reviewed with the recording and action trace.
2	2	agent_observed_resource_balance	134	Automated playthrough observed resource balance issue [RESOURCE_BALANCE] Starting ammo starvation. At tick 133, position (-636, 370): Player has 0 ammo and no immediate pickups. Player impact: Unable to engage enemies or defend against melee pressure. Severity: major.
2	2	agent_observed_progression	385	Automated playthrough observed progression issue [PROGRESSION] Navigation stuck loop. At tick 382, position (-924.89, 365.52): Player is repeatedly stuck trying to path to pickups. Player impact: Cannot reach weapons to defend against spawned enemies. Severity: major.
2	2	agent_observed_geometry	393	Automated playthrough observed geometry issue [GEOMETRY] Chainsaw collision. At tick 390, position (-892.35, 389.41): Player is unable to navigate to reachable Chainsaw (id 5, 25) due to wall collision. Player impact: Cannot arm melee weapon for defense. Severity: major.
3	3	unreachable_secret	603	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Central corridor and item storage zones:** high Collision geometry prevents the automated playthrough from accessing essential weaponry, leading to total progression failure.

Activity Log Summary

1) Static analysis phase, 2) Navigation attempt (ticks 15-126), 3) Combat/Resource acquisition phase (ticks 134-523), 4) Defect detection phase, 5) Report generation.

The automated playthrough did not engage in combat as planned due to a total lack of usable ammunition and unreachable weaponry.

Limitations And Recommendations

The automated playthrough did not test weapon pickup functionality due to geometry collision barriers, preventing the automated playthrough from testing combat mechanics fully.

Good Quality Areas

- **Initial spawn zone:** The automated playthrough was able to collect initial items (CellPack/Chainsaw) indicating that the starting area is correctly navigable.

Evidence Appendix

Notable Event Trace

Tick	Event	Position	HP	MCP Action	Stop reason
126	stuck	-646.6, 354.1	100	take_action	tics_complete
136	stuck	-633.5, 376.2	100	take_action	tics_complete
240	stuck	-668.2, -119.6	100	take_action	tics_complete
328	item_pickup	-1139.2, 97.1	100	move_to	arrived
330	stuck	-1143.2, 102.2	100	take_action	tics_complete
379	item_pickup	-930.6, 361.3	100	move_to	arrived
381	stuck	-924.9, 365.5	100	take_action	tics_complete
468	stuck	-1198.7, 437.8	100	take_action	tics_complete
482	stuck	-1208.1, 440.6	100	move_to	enemy_nearby
523	damage_taken	-1011.0, 273.3	94	retreat	complete
603	item_pickup	-592.0, 40.7	94	retreat	complete

LLM/MCP Decision Trace

#0 ticks 14 -> 15 tool explore stop item_found I am initiating exploration toward visible resources while keeping distance from the barrel clusters and potential threat zones.
#1 ticks 16 -> 40 tool move_to stop arrived I am gathering the visible CellPack to secure resources before engaging the visible enemy.
#2 ticks 41 -> 63 tool move_to stop arrived I am gathering the nearby Chainsaw weapon to improve my close-quarters defense capability, as no threats are currently visible and I need to build up resources.
#3 ticks 64 -> 103 tool move_to stop pickup_not_collected I am picking up the visible Chainsaw to increase my resource pool before venturing out to investigate the enemy positions.
#4 ticks 104 -> 124 tool move_to stop arrived I am currently out of projectile ammo and have secured the chainsaw. I will proceed to the nearby ShellBox to gather ammunition before engaging the surrounding enemies.
#5 ticks 125 -> 126 tool take_action stop tics_complete Object 34 was already reached or collected in this run, so I will not target it again. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#6 | ticks 127 -> 128 | tool explore | stop enemy_spotted

I am currently out of primary ammunition and need to explore for further resources or weapon pickups, as the current area is empty of reachable ammo. I will move north to explore and locate items.

#7 | ticks 129 -> 130 | tool explore | stop enemy_spotted

I am currently out of primary ammunition and need to explore for further resources or weapon pickups, as the current area is empty of reachable ammo. I will move north to explore and locate items.

#8 | ticks 131 -> 132 | tool explore | stop enemy_spotted

I am currently out of primary ammunition and need to explore to find a weapon or ammo pickup. I am moving north to explore new areas.

#9 | ticks 133 -> 134 | tool explore | stop enemy_spotted

I am currently out of ammunition and the surrounding area provides no immediate resources. I am initiating an exploration move to locate necessary pickups before engaging the hostile demons and imps.

#10 | ticks 135 -> 136 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#11 | ticks 137 -> 138 | tool take_action | stop tics_complete

I am out of primary weapon ammo and need to locate a weapon or supplies. I will turn to navigate toward potential item locations not yet fully cleared in my navigation path.

#12 | ticks 139 -> 141 | tool take_action | stop tics_complete

I am currently out of primary ammunition and need to find a weapon or ammo pickup to progress. Since previous exploration attempts have been hampered by navigation blocks, I am performing a small turn to reorient toward unexplored areas.

#13 | ticks 142 -> 180 | tool move_to | stop pickup_not_collected

I am currently out of ammunition and need a weapon. I am targeting the visible Chainsaw to secure a method of defense.

#14 | ticks 181 -> 217 | tool move_to | stop pickup_not_collected

I am currently out of ammunition and the Chainsaw (ID 25) is clearly visible. I am moving to pick up the Chainsaw to allow for combat progression.

#15 | ticks 218 -> 238 | tool move_to | stop arrived

I am currently out of ammunition and the nearest ammo pickup (ClipBox 35) is visible. I will move to collect this to enable potential combat.

#16 | ticks 239 -> 240 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#17 | ticks 241 -> 307 | tool move_to | stop pickup_not_collected

I am currently out of effective ammunition (0 ammo in slot 1). I will attempt to collect the Chainsaw (ID 24) to establish a means of defense before engaging the visible Imp.

#18 | ticks 308 -> 328 | tool move_to | stop arrived

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am switching to available pickup Infrared first.

#19 | ticks 329 -> 330 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#20 | ticks 331 -> 379 | tool move_to | stop arrived

I am currently out of ammunition and the surrounding area provides no weapons or ammo. I will navigate towards the visible Infrared item to maximize visibility and then search for reachable resources or map exits.

#21 | ticks 380 -> 381 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#22 | ticks 382 -> 385 | tool take_action | stop tics_complete

I am currently out of ammo and previous attempts to navigate to pickups (Chainsaw) resulted in stuck states. I will rotate to scan for other potential paths or items to break the current stagnation.

#23 | ticks 386 -> 389 | tool take_action | stop tics_complete

I am currently out of ammo and previous attempts to navigate to the visible Chainsaw failed due to collision geometry. I will turn to face the unexplored north sector to search for accessible ammo or exit paths.

#24 | ticks 390 -> 393 | tool take_action | stop tics_complete

I am currently out of ammo and previous attempts to navigate to the visible Chainsaw failed due to collision geometry. I will turn to face the unexplored north sector to search for accessible ammo or exit paths.

#25 | ticks 394 -> 401 | tool take_action | stop tics_complete

I am currently out of ammo and previous attempts to navigate to the visible Chainsaw failed due to collision geometry. I will turn to face the unexplored north sector to search for accessible ammo or exit paths.

#26 | ticks 402 -> 438 | tool explore | stop enemy_spotted

I am currently out of primary ammo and all reachable weapon/ammo pickups (Chainsaws) are blocked by collision geometry. I will use the exploration tool to find a different path or secret to bypass the geometry constraints.

#27 | ticks 439 -> 446 | tool take_action | stop tics_complete

I am completely out of ammo and all reachable weapon pickups are blocked by collision geometry, causing a progression stall. I will probe the environment with a small movement to see if I can navigate around the invisible barriers near the center.

#28 | ticks 447 -> 454 | tool take_action | stop tics_complete

I am unable to reach ammo or weapons due to persistent collision barriers around the central area. I will attempt to move forward to test if the wall geometry allows passage, confirming the progression stall.

#29 | ticks 455 -> 462 | tool take_action | stop tics_complete

I am currently out of ammo and previous attempts to navigate to the visible Chainsaws were blocked by collision geometry. I will attempt a precise movement to probe the nearby wall to confirm the progression stall.

#30 | ticks 463 -> 464 | tool take_action | stop tics_complete

I am completely out of ammunition and all reachable weapon/ammo pickups (Chainsaws) are confirmed to be behind blocking collision geometry, preventing progression. I am marking this as a progression failure.

#31 | ticks 465 -> 466 | tool take_action | stop tics_complete

I am completely out of ammunition and the reachable weapon pickups (Chainsaws) are behind collision geometry that I cannot traverse. Since I cannot engage the nearby Imp without ammo or reach the weapons, this is a confirmed progression failure.

#32 | ticks 467 -> 468 | tool take_action | stop tics_complete

Progress has not changed across repeated lockstep decisions, so I am forcing a bounded QA recovery probe. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#33 | ticks 469 -> 470 | tool take_action | stop tics_complete

I am currently out of ammo and all reachable weapon/ammo pickups are confirmed to be behind blocking geometry, rendering the map non-progressable. I am formally reporting the progression blockage.

#34 | ticks 471 -> 474 | tool take_action | stop tics_complete

I am out of ammunition and all reachable weapon/ammo pickups (Chainsaws) are confirmed to be behind blocking collision geometry, rendering the map non-progressable. I am formally reporting the progression blockage.

#35 | ticks 475 -> 478 | tool take_action | stop tics_complete

I am completely out of ammunition and all visible potential weapons (Chainsaws) are behind collision barriers that I cannot traverse. The map is non-progressable as no enemies can be defeated to clear space or trigger lines.

#36 | ticks 479 -> 480 | tool aim_and_shoot | stop out_of_ammo

Model response failed or was rate limited; using deterministic fallback. Visible DoomImp selected for combat.

#37 | ticks 481 -> 482 | tool move_to | stop enemy_nearby

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am switching to available pickup Infrared first.

#38 | ticks 483 -> 484 | tool take_action | stop tics_complete

I am completely out of ammunition and all visible potential weapons are behind collision barriers that I cannot traverse. The map is non-progressable as no enemies can be defeated to clear space or trigger lines, and navigation is currently blocked.

#39 | ticks 485 -> 523 | tool retreat | stop complete

I am trapped in a confined space with a DoomImp at close range, out of ammunition, and unable to reach any functional weapons or pickups due to collision barriers. This is a critical progression blockage.